

Indraprastha Institute of Information Technology, Delhi

IIIT-Delhi - Department of Computer Science & Engineering

Discrete Mathematics - Unit Notes

Course Resource Package

IIIT-Delhi CSE111 Lecture Notes:

- Pigeonhole Principle: If $k+1$ objects placed in k boxes, at least one box has ≥ 2 objects.
- Equivalence relation is reflexive, symmetric, and transitive.
- Handshaking Lemma: Sum of degrees of all vertices is twice the number of edges.
- Planar Graph: Satisfies Euler's formula: $V - E + F = 2$.
- Lagrange's Theorem: Order of a subgroup divides the order of the group.